API STD*526 95 ■ 0732290 0545992 935 ■

Flanged Steel Pressure Relief Valves

API STANDARD 526
FOURTH EDITION, JUNE 1995

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



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Flanged Steel Pressure Relief Valves

Manufacturing, Distribution and Marketing Department

API STANDARD 526
FOURTH EDITION, JUNE 1995

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FOREWORD

This standard is a purchase specification for flanged steel pressure relief valves including direct spring loaded pressure relief valves and pilot-operated pressure relief valves.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or incorporating sections thereof into another complete standard.

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Flanged Steel Pressure Relief Valves

SECTION 1—GENERAL

1.1 Scope

This standard is a purchase specification for flanged steel pressure relief valves. Basic requirements are given for direct spring-loaded pressure relief valves and pilot-operated pressure relief valves as follows:

a. Orifice designation and area.
b. Valve size and pressure rating, inlet and outlet.
c. Materials.
d. Pressure-temperature limits.
e. Center-to-face dimensions, inlet and outlet.

For the convenience of the purchaser, a sample specification sheet is given in Appendix A.

Nameplate nomenclature and requirements for stamping are detailed in Appendix B.

1.2 Referenced Publications

The latest edition or revision of the following standards, codes, and specifications shall, to the extent specified, form a part of this standard.

API
- RP 520 *Sizing, Selection, and Installation of Pressure Relieving Devices in Refineries*, Parts I and II
- Std 527 *Seat Tightness of Pressure Relief Valves*

ASME[1]

Boiler and Pressure Vessel Code
- Section VIII, *Pressure Vessels*, Division 1 and 2
- Section II—*Material Specifications*
  - SA 216 *Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service*
  - SA 217 *Martensitic Stainless Steel and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service*
  - SA 351 *Austenitic Steel Castings for High-Temperature Service*

ASME/ANSI[2]
- B16.5 *Pipe Flanges and Flanged Fittings*
- B16.34 *Valves—Flanged, Threaded, and Welding End*

NACE Int'l[3]
- MR0175-90 *Sulfide Stress Cracking Resistant Metallic Materials for Oil Field Equipment*

1.3 Definitions

Pressure relief valve terminology is defined in API Recommended Practice 520, Part I.

1.4 Responsibility

The purchaser is primarily responsible for the following:

a. Selecting the type of pressure relief valve and the required pressure-temperature ratings.
b. Specifying materials that will satisfactorily resist corrosion from the process fluid and environmental conditions.
c. Selecting the minimum required orifice area based upon relieving conditions derived from full knowledge of the pressure-relieving system and the requirements of the applicable codes and regulations.
d. Providing data for sizing and selection.

The manufacturer is primarily responsible for the following:

a. Designing and manufacturing pressure relief valves to satisfy the requirements of this standard and the purchaser's specification.
b. Publishing relieving capacities based upon certified test data.
c. Advising the purchaser of any nonconformance to the purchaser's specification.

1.5 Conflicting Requirements

Whenever the information included on the purchaser's specification sheet or purchase order conflicts with the provisions of this standard, the purchaser's specification sheet or purchase order shall govern. Where conflicting requirements exist, the manufacturer shall call them to the attention of the purchaser.

1.6 Orifice Areas and Designations

The standard effective orifice areas and the corresponding letter designations are listed in Table 1. These effective areas are valid only when used with the sizing equations contained in API RP 520, Part I.

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017
[2]American National Standards Institute, 11 West 42nd Street, New York, NY 10036
[3]NACE International, P.O. Box 218340, Houston, Texas 77218

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Table 1—Standard Effective Orifice Areas and Letter Designations

Designation	Effective Orifice Area (square inches)
D	0.110
E	0.196
F	0.307
G	0.503
H	0.785
J	1.287
K	1.838
L	2.853
M	3.60
N	4.34
P	6.38
Q	11.05
R	16.0
T	26.0

SECTION 2—DESIGN

2.1 General

Pressure relief valves discussed in this standard shall be designed and manufactured in accordance with the applicable requirements of Section VIII of the ASME *Boiler and Pressure Vessel Code* for pressure relief devices.

2.2 Determination of Orifice Area

The minimum required effective orifice area shall be determined in accordance with API RP 520, Part I.

2.3 Valve Selection

For valves described in this standard, inlet and outlet flange sizes and pressure-temperature ratings shall conform to the data in Tables 2 through 15 for spring-loaded valves and Tables 16 through 29 for pilot-operated valves. Inlet and outlet pressure limits are governed by the flange pressure/temperature limits or by the manufacturer's design limits, whichever is lower.

2.4 Dimensions

Center-to-face dimensions shall be in accordance with Tables 2 through 15 for spring-loaded valves and Tables 16 through 29 for pilot-operated valves, with tolerances of plus or minus 1/16 inch for valve inlet sizes up to and including 4 inches, and plus or minus 1/8 inch for valve inlet sizes larger than 4 inches.

Flange facings and dimensions shall be in accordance with ASME/ANSI B16.5.

For some valve designs, the inlet raised face height may substantially exceed the nominal dimension specified in ASME/ANSI B16.5. Consult the manufacturer for exact dimensions.

2.5 Lifting Levers

Lifting levers shall be provided when required by the ASME *Boiler and Pressure Vessel Code* or when designated on the purchaser's specification sheet.

As allowed by code for pilot-operated pressure relief valves, means in lieu of lifting levers may be specified for connecting and applying adequate pressure to the pilot to verify that the moving parts critical to proper operation are free to move.

2.6 Special Construction Features

Construction features beyond the scope of this standard shall be provided as agreed upon by the manufacturer and the purchaser and shall be designated on the drawings provided to the purchaser.

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SECTION 3—MATERIAL

3.1 General

Materials generally used for construction are covered in this section. For special corrosion problems and applications beyond the pressure-temperature limits of this standard, construction materials shall be those agreed upon by the manufacturer and the purchaser.

3.2 Spring-Loaded Pressure Relief Valves

The body, bonnet, and spring materials shall be in accordance with Tables 2 through 15 for the required temperature range. The body and bonnet may be of different materials but must meet the minimum pressure-temperature requirement. The body and bonnet materials shall be equivalent to or better than the following types and grades:

Carbon steel	ASME SA 216, Grade WCB
Chromium molybdenum steel	ASME SA 217, Grade WC6
Austenitic stainless steel	ASME SA 351, Grade CF8M

Material for the internal parts of the valve shall be in accordance with the manufacturer's standards for the temperature and service or as indicated on the purchaser's specification sheet.

3.3 Pilot-Operated Pressure Relief Valves

The body material shall be in accordance with Tables 16 through 29 for the required temperature range. The body material shall be equivalent to or better than the following types and grades:

Carbon steel	ASME SA 216, Grade WCB
Austenitic stainless steel	ASME SA 351, Grade CF8M

Material for the pilot and internal parts of the valve shall be in accordance with the manufacturer's standards for the temperature and service or as indicated on the purchaser's specification sheet.

SECTION 4—INSPECTION AND SHOP TESTS

4.1 Inspection

The purchaser reserves the right to witness the shop tests and inspect the valves in the manufacturer's plant to the extent specified on the purchase order.

4.2 Set Pressure Test

All pressure relief valves shall be adjusted to the specified set pressure in accordance with the ASME *Boiler and Pressure Vessel Code*, the manufacturer's standard practice as published, or as designated by the purchaser. All set pressure adjustments shall be sealed.

4.3 Seat Leakage Test

All pressure relief valves shall be seat leakage tested in accordance with API Standard 527, or as agreed upon by the manufacturer and the purchaser.

SECTION 5—IDENTIFICATION AND PREPARATION FOR SHIPMENT

5.1 Identification

Each pressure relief valve shall have a corrosion-resistant nameplate permanently attached to the body or bonnet. This nameplate shall be stamped with the data specified in Appendix B. Pilot-operated pressure relief valves shall have an additional nameplate permanently attached to the pilot. The pilot nameplate shall be stamped with the manufacturer's name, pilot type, set pressure, and serial number.

Each pressure relief valve shall be tagged with the purchaser's valve number or other identification as specified on the purchase order. The data may be stamped on the nameplate or on a separate corrosion-resistant tag that is permanently attached to the valve.

5.2 Preparation for Shipment

Each pressure relief valve shall be prepared for shipment as follows:

a. After test and inspection, all exterior surfaces, except flange facings, shall be painted as agreed upon by the manufacturer and the purchaser. Corrosion-resistant materials need not be painted. Flange facings shall be coated with a suitable corrosion inhibitor.

b. Threaded openings shall be plugged with suitable protective devices. Temporary plugs should be readily distinguishable from permanent metal plugs.

c. Inlet and outlet flanges shall be protected to prevent damage from or entrance of foreign material during shipment.

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Table 2—Spring-Loaded Pressure Relief Valves
"D" Orifice (Effective Orifice Area = 0.110 square inch)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig)									
					Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	1D2	150	150			285	185			285	230	4-⅛	4-½
		1D2	300	150			285	285			285	230	4-⅛	4-½
		1D2	300	150			740	615			285	230	4-⅛	4-½
		1D2	600	150			1480	1235			285	230	4-⅛	4-½
		1-½ D2	900	300			2220	1845			600	500	4-⅛	5-½
		1-½ D2	1500	300			3705	3080			600	500	4-⅛	5-½
		1-½ D3	2500	300			6000	5135			740	500	5-½	7
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	1D2	150	150				185	80		285	230	4-⅛	4-½
		1D2	300	150				285	285		285	230	4-⅛	4-½
		1D2	300	150				615	410		285	230	4-⅛	4-½
		1D2	600	150				1235	825		285	230	4-⅛	4-½
		1-½ D2	900	300				1845	1235		600	500	4-⅛	5-½
		1-½ D2	1500	300				3080	2060		600	500	4-⅛	5-½
		1-½ D3	2500	300				5135	3430		740	500	5-½	7
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	1D2	300	150					510	225	285	230	4-⅛	4-½
		1D2	600	150					1015	445	285	230	4-⅛	4-½
		1-½ D2	900	300					1525	670	600	500	4-⅛	5-½
		1-½ D2	1500	300					2540	1115	600	500	4-⅛	5-½
		1-½ D3	2500	300					4230	1860	740	500	5-½	7
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	1D2	150	150		275					275	230	4-⅛	4-½
		1D2	300	150		275					275	230	4-⅛	4-½
		1D2	300	150		720					275	230	4-⅛	4-½
		1D2	600	150		1440					275	230	4-⅛	4-½
		1-½ D2	900	300		2160					600	500	4-⅛	5-½
		1-½ D2	1500	300		3600					600	500	4-⅛	5-½
		1-½ D3	2500	300		6000					720	500	5-½	7
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	1D2	150	150	275						275	230	4-⅛	4-½
		1D2	300	150	275						275	230	4-⅛	4-½
		1D2	300	150	720						275	230	4-⅛	4-½
		1D2	600	150	1440						275	230	4-⅛	4-½
		1-½ D2	900	300	2160						600	500	4-⅛	5-½
		1-½ D2	1500	300	3600						600	500	4-⅛	5-½
		1-½ D3	2500	300	4000						720	500	5-½	7

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 3—Spring-Loaded Pressure Relief Valves
"E" Orifice (Effective Orifice Area = 0.196 square inch)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	1E2	150	150			285	185			285	230	4-⅛	4-½
		1E2	300	150			285	285			285	230	4-⅛	4-½
		1E2	300	150			740	615			285	230	4-⅛	4-½
		1E2	600	150			1480	1235			285	230	4-⅛	4-½
		1-½ E2	900	300			2220	1845			600	500	4-⅛	5-½
		1-½ E2	1500	300			3705	3080			600	500	4-⅛	5-½
		1-½ E3	2500	300			6000	5135			740	500	5-½	7
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	1E2	150	150				185	80		285	230	4-⅛	4-½
		1E2	300	150				285	285		285	230	4-⅛	4-½
		1E2	300	150				615	410		285	230	4-⅛	4-½
		1E2	600	150				1235	825		285	230	4-⅛	4-½
		1-½ E2	900	300				1845	1235		600	500	4-⅛	5-½
		1-½ E2	1500	300				3080	2060		600	500	4-⅛	5-½
		1-½ E3	2500	300				5135	3430		740	500	5-½	7
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	1E2	300	150					510	225	285	230	4-⅛	4-½
		1E2	600	150					1015	445	285	230	4-⅛	4-½
		1-½ E2	900	300					1525	670	600	500	4-⅛	5-½
		1-½ E2	1500	300					2540	1115	600	500	4-⅛	5-½
		1-½ E3	2500	300					4230	1860	740	500	5-½	7
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	1E2	150	150		275					275	230	4-⅛	4-½
		1E2	300	150		275					275	230	4-⅛	4-½
		1E2	300	150		720					275	230	4-⅛	4-½
		1E2	600	150		1440					275	230	4-⅛	4-½
		1-½ E2	900	300		2160					600	500	4-⅛	5-½
		1-½ E2	1500	300		3600					600	500	4-⅛	5-½
		1-½ E3	2500	300		6000					720	500	5-½	7
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	1E2	150	150	275						275	230	4-⅛	4-½
		1E2	300	150	275						275	230	4-⅛	4-½
		1E2	300	150	720						275	230	4-⅛	4-½
		1E2	600	150	1440						275	230	4-⅛	4-½
		1-½ E2	900	300	2160						600	500	4-⅛	5-½
		1-½ E2	1500	300	3600						600	500	4-⅛	5-½
		1-½ E3	2500	300	4000						720	500	5-½	7

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 4—Spring-Loaded Pressure Relief Valves "°F" Orifice (Effective Orifice Area = 0.307 square inch)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	1-½ F2	150	150			285	185			285	230	4-⅞	4-¼
		1-½ F2	300	150			285	285			285	230	4-⅞	4-¼
		1-½ F2	300	150			740	615			285	230	4-⅞	6
		1-½ F2	600	150			1480	1235			285	230	4-⅞	6
		1-½ F3	900	300			2220	1845			740	500	4-⅞	6-½
		1-½ F3	1500	300			3705	3080			740	500	4-⅞	6-½
		1-½ F3	2500	300			5000	5000			740	500	5-½	7
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	1-½ F2	150	150				185	80		285	230	4-⅞	4-¼
		1-½ F2	300	150				285	285		285	230	4-⅞	4-¼
		1-½ F2	300	150				615	410		285	230	4-⅞	6
		1-½ F2	600	150				1235	825		285	230	4-⅞	6
		1-½ F3	900	300				1845	1235		740	500	4-⅞	6-½
		1-½ F3	1500	300				3080	2060		740	500	4-⅞	6-½
		1-½ F3	2500	300				5000	3430		740	500	5-½	7
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	1-½F2	300	150					510	225	285	230	4-⅞	6
		1-½F2	600	150					1015	445	285	230	4-⅞	6
		1-½ F3	900	300					1525	670	740	500	4-⅞	6-½
		1-½ F3	1500	300					2540	1115	740	500	4-⅞	6-½
		1-½ F3	2500	300					4230	1860	740	500	5-½	7
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	1-½ F2	150	150		275					275	230	4-⅞	4-¼
		1-½ F2	300	150		275					275	230	4-⅞⅛	4-¼
		1-½ F2	300	150		720					275	230	4-⅞	6
		1-½ F2	600	150		1440					275	230	4-⅞	6
		1-½ F3	900	300		2160					720	500	4-⅞	6-½
		1-½ F3	1500	300		3600					720	500	4-⅞	6-½
		1-½ F3	2500	300		5000					720	500	5-½	7
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	1-½ F2	150	150	275						275	230	4-⅞	4-¼
		1-½ F2	300	150	275						275	230	4-⅞	4-¼
		1-½ F2	300	150	720						275	230	4-⅞	6
		1-½ F2	600	150	1440						275	230	4-⅞	6
		1-½ F3	900	300	2160						720	500	4-⅞	6-½
		1-½ F3	1500	300	2200						720	500	4-⅞	6-½
		1-½ F3	2500	300	3400						720	500	5-½	7

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 5—Spring-Loaded Pressure Relief Valves
"G" Orifice (Effective Orifice Area = 0.503 square inch)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	1-½ G3	150	150			285	185			285	230	4-⅞	4-¾
		1-½ G3	300	150			285	285			285	230	4-⅞	4-¾
		1-½ G3	300	150			740	615			285	230	4-⅞	6
		1-½ G3	600	150			1480	1235			285	230	4-⅞	6
		1-½ G3	900	300			2220	1845			740	470	4-⅞	6-½
		2G3	1500	300			3705	3080			740	470	6-⅛	6-¾
		2G3	2500	300			3705	3705			740	470	6-⅛	6-¾
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	1-½ G3	150	150				185	80		285	230	4-⅞	4-¾
		1-½ G3	300	150				285	285		285	230	4-⅞	4-¾
		1-½ G3	300	150				615	410		285	230	4-⅞	6
		1-½ G3	600	150				1235	825		285	230	4-⅞	6
		1-½ G3	900	300				1845	1235		740	470	4-⅞	6-1/2
		2G3	1500	300				3080	2060		740	470	6-⅛	6-¾
		2G3	2500	300				3705	3430		740	470	6-⅛	6-¾
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	1-½ G3	300	150					510	225	285	230	4-⅞	6
		1-½ G3	600	150					1015	445	285	230	4-⅞	6
		1-½ G3	900	300					1525	670	740	470	4-⅞	6-½
		2G3	1500	300					2540	1115	740	470	6-⅛	6-¾
		2G3	2500	300					3705	1860	740	470	6-⅛	6-¾
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	1-½ G3	150	150		275					275	230	4-⅞	4-¾
		1-½ G3	300	150		275					275	230	4-⅞	4-¾
		1-½ G3	300	150		720					275	230	4-⅞	6
		1-½ G3	600	150		1440					275	230	4-⅞	6
		1-½ G3	900	300		2160					720	470	4-⅞	6-½
		2G3	1500	300		3600					720	470	6-⅛	6-¾
		2G3	2500	300		3600					720	470	6-⅛	6-¾
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	1-½ G3	150	150	275						275	230	4-⅞	4-¾
		1-½ G3	300	150	275						275	230	4-⅞	4-¾
		1-½ G3	300	150	720						275	230	4-⅞	6
		1-½ G3	600	150	1440						275	230	4-⅞	6
		1-½ G3	900	300	1600						720	470	4-⅞	6-½
		2G3	1500	300	2450						720	470	6-⅛	6-¾
		2G3	2500	300	2600						720	470	6-⅛	6-¾

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 6—Spring-Loaded Pressure Relief Valves
"H" Orifice (Effective Orifice Area = 0.785 square inch)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	1-½ H3	150	150			285	185			285	230	5-⅛	4-⅞
		1-½ H3	300	150			285	285			285	230	5-⅛	4-⅞
		2H3	300	150			740	615			285	230	5-⅛	4-⅞
		2H3	600	150			1480	1235			285	230	6-1/16	6-⅜
		2H3	900	150			2220	1845			285	230	6-1/16	6-⅜
		2H3	1500	300			2750	2750			740	415	6-1/16	6-⅜
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	1-½ H3	150	150				185	80		285	230	5-⅛	4-⅞
		1-½ H3	300	150				285	285		285	230	5-⅛	4-⅞
		2H3	300	150				615	410		285	230	5-⅛	4-⅞
		2H3	600	150				1235	825		285	230	6-1/16	6-⅜
		2H3	900	150				1845	1235		285	230	6-1/16	6-⅜
		2H3	1500	300				2750	2060		740	415	6-1/16	6-⅜
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	2H3	300	150					510	225	285	230	5-⅛	4-⅞
		2H3	600	150					815	445	285	230	5-⅛	4-⅞
		2H3	900	150					1225	670	285	230	6-1/16	6-⅜
		2H3	1500	300					2040	1115	740	415	6-1/16	6-⅜
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	1-½ H3	150	150		275					275	230	5-⅛	4-⅞
		1-½ H3	300	150		275					275	230	5-⅛	4-⅞
		2H3	300	150		720					275	230	5-⅛	4-⅞
		2H3	600	150		1440					275	230	6-1/16	6-⅜
		2H3	900	150		2160					275	230	6-1/16	6-⅜
		2H3	1500	300		2750					720	415	6-1/16	6-⅜
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	1-½ H3	150	150	275						275	230	5-⅛	4-⅞
		1-½ H3	300	150	275						275	230	5-⅛	4-⅞
		2H3	300	150	720						275	230	5-⅛	4-⅞
		2H3	600	150	1440						275	230	6-1/16	6-⅜
		2H3	900	150	1485						275	230	6-1/16	6-⅜
		2H3	1500	300	1600						720	415	6-1/16	6-⅜

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 7—Spring-Loaded Pressure Relief Valves
"J" Orifice (Effective Orifice Area = 1.287 square inches)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig)									
					Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	2J3	150	150			285	185			285	230	5-⅜	4-⅞
		2J3	300	150			285	285			285	230	5-⅜	4-⅞
		3J4	300	150			740	615			285	230	7-¼	7-⅛
		3J4	600	150			1480	1235			285	230	7-¼	7-⅛
		3J4	900	150			2220	1845			285	230	7-¼	7-⅛
		3J4	1500	300			2700	2700			600	230	7-¼	7-⅛
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	2J3	150	150				185	80		285	230	5-⅜	4-⅞
		2J3	300	150				285	285		285	230	5-⅜	4-⅞
		3J4	300	150				615	410		285	230	7-¼	7-⅛
		3J4	600	150				1235	825		285	230	7-¼	7-⅛
		3J4	900	150				1845	1235		285	230	7-¼	7-⅛
		3J4	1500	300				2700	2060		600	230	7-¼	7-⅛
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	3J4	300	150					510	225	285	230	7-¼	7-⅛
		3J4	600	150					815	445	285	230	7-¼	7-⅛
		3J4	900	150					1225	670	285	230	7-¼	7-⅛
		3J4	1500	300					2040	1115	600	230	7-¼	7-⅛
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	2J3	150	150		275					275	230	5-⅜	4-⅞
		2J3	300	150		275					275	230	5-⅜	4-⅞
		3J4	300	150		720					275	230	7-¼	7-⅛
		3J4	600	150		1440					275	230	7-¼	7-⅛
		3J4	900	150		2160					275	230	7-¼	7-⅛
		3J4	1500	300		2700					600	230	7-¼	7-⅛
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	2J3	150	150	275						275	230	5-⅜	4-⅞
		2J3	300	150	275						275	230	5-⅜	4-⅞
		3J4	300	150	500						275	230	7-¼	7-⅛
		3J4	600	150	625						275	230	7-¼	7-⅛
		3J4	900	150	800						275	230	7-¼	7-⅛
		3J4	1500	300	800						600	230	7-¼	7-⅛

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 8—Spring-Loaded Pressure Relief Valves "K" Orifice (Effective Orifice Area = 1.838 square inches)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	3K4	150	150			285	185			285	150	6-⅛	6-⅜
		3K4	300	150			285	285			285	150	6-⅛	6-⅜
		3K4	300	150			740	615			285	150	6-⅛	6-⅜
		3K4	600	150			1480	1235			285	200	7-¼	7-⅛
		3K6	900	150			2220	1845			285	200	7-13/16	8-½
		3K6	1500	300			2220	2220			600	200	7-¼	8-½
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	3K4	150	150				185	80		285	150	6-⅛	6-⅜
		3K4	300	150				285	285		285	150	6-⅛	6-⅜
		3K4	300	150				615	410		285	150	6-⅛	6-⅜
		3K4	600	150				1235	825		285	200	7-¼	7-⅛
		3K6	900	150				1845	1235		285	200	7-13/16	8-½
		3K6	1500	300				2220	2060		600	200	7-¼	8-½
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	3K4	300	150					510	225	285	150	6-⅛	6-⅜
		3K4	600	150					815	445	285	200	6-⅛	6-⅜
		3K6	900	150					1225	670	285	200	7-¼	7-⅛
		3K6	1500	300					2040	1115	600	200	7-¼	8-½
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	3K4	150	150		275					275	150	6-⅛	6-⅜
		3K4	300	150		275					275	150	6-⅛	6-⅜
		3K4	300	150		720					275	150	6-⅛	6-⅜
		3K4	600	150		1440					275	200	7-¼	7-⅛
		3K6	900	150		2160					275	200	7-13/16	8-½
		3K6	1500	300		2220					600	200	7-¼	8-½
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	3K4	150	150	275						275	150	6-⅛	6-⅜
		3K4	300	150	275						275	150	6-⅛	6-⅜
		3K4	300	150	525						275	150	6-⅛	6-⅜
		3K4	600	150	600						275	200	7-¼	7-⅛
		3K6	900	150	600						275	200	7-13/16	8-½
		3K6	1500	300	750						600	200	7-¼	8-½

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 9—Spring-Loaded Pressure Relief Valves
"L" Orifice (Effective Orifice Area = 2.853 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	3L4	150	150			285	185			285	100	6-¼	6-½
		3L4	300	150			285	285			285	100	6-¼	6-½
		4L6	300	150			740	615			285	170	7-1/16	7-⅛
		4L6	600	150			1000	1000			285	170	7-1/16	8
		4L6	900	150			1500	1500			285	170	7-¾	8-¼
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	3L4	150	150				185	80		285	100	6-¼	6-½
		3L4	300	150				285	285		285	100	6-¼	6-½
		4L6	300	150				615	410		285	170	7-1/16	7-⅛
		4L6	600	150				1000	825		285	170	7-1/16	8
		4L6	900	150				1500	1235		285	170	7-¾	8-¼
		4L6	1500	150				1500	1500		285	170	7-¾	8-¼
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	4L6	300	150					510	225	285	170	7-1/16	7-⅛
		4L6	600	150					1000	445	285	170	7-⅛	8
		4L6	900	150					1500	670	285	170	7-¼	8-¼
		4L6	1500	150					1500	1115	285	170	7-¼	8-¼
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	3L4	150	150		275					275	100	6-⅛	6-½
		3L4	300	150		275					275	100-	6-⅛	6-½
		4L6	300	150		720					275	170	7-1/16	7-⅛
		4L6	600	150		1000					275	170	7-1/16	8
		4L6	900	150		1500					275	170	7-¾	8-¼
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	3L4	150	150	275						275	100	6-⅛	6-½
		3L4	300	150	275						275	100	6-⅛	6-½
		4L6	300	150	535						275	170	7-1/16	7-⅛
		4L6	600	150	535						275	170	7-1/16	8
		4L6	900	150	700						275	170	7-¾	8-¼

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 10—Spring-Loaded Pressure Relief Valves "M" Orifice (Effective Orifice Area = 3.60 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	4M6	150	150			285	185			285	80	7	7-¼
		4M6	300	150			285	285			285	80	7	7-¼
		4M6	300	150			740	615			285	160	7	7-¼
		4M6	600	150			1100	1100			285	160	7	8
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	4M6	150	150				185	80		285	80	7	7-¼
		4M6	300	150				285	285		285	80	7	7-¼
		4M6	300	150				615	410		285	160	7	7-¼
		4M6	600	150				1100	825		285	160	7	8
		4M6	900	150				1100	1100		285	160	7-¾	8-¾
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	4M6	300	150					510	225	285	160	7	7-1/4
		4M6	600	150					1015	445	285	160	7	8
		4M6	900	150					1100	670	285	160	7-¾	8-¾
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	4M6	150	150		275					275	80	7	7-¼
		4M6	300	150		275					275	80	7	7-¼
		4M6	300	150		720					275	160	7	7-¼
		4M6	600	150		1000					275	160	7	8
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	4M6	150	150	275						275	80	7	7-¼
		4M6	300	150	275						275	80	7	7-¼
		4M6	300	150	525						275	160	7	7-¼
		4M6	600	150	600						275	160	7	8

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 11—Spring-Loaded Pressure Relief Valves "N" Orifice (Effective Orifice Area = 4.34 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	4N6	150	150			285	185			285	80	7-¾	8-¼
		4N6	300	150			285	285			285	80	7-¾	8-¼
		4N6	300	150			740	615			285	160	7-¾	8-¼
		4N6	600	150			1000	1000			285	160	7-¾	8-¾
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	4N6	150	150				185	80		285	80	7-¾	8-¼
		4N6	300	150				285	285		285	80	7-¾	8-¼
		4N6	300	150				615	410		285	160	7-¾	8-¼
		4N6	600	150				1000	825		285	160	7-¾	8-¾
		4N6	900	150				1000	1000		285	160	7-¾	8-¾
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	4N6	300	150					510	225	285	160	7-¾	8-¼
		4N6	600	150					1000	445	285	160	7-¾	8-¾
		4N6	900	150					1000	670	285	160	7-¾	8-¾
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	4N6	150	150		275					275	80	7-¾	8-¼
		4N6	300	150		275					275	80	7-¾	8-¼
		4N6	300	150		720					275	160	7-¾	8-¼
		4N6	600	150		1000					275	160	7-¾	8-¾
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	4N6	150	150	275						275	80	7-¾	8-¼
		4N6	300	150	275						275	80	7-¾	8-¼
		4N6	300	150	450						275	160	7-¾	8-¼
		4N6	600	150	500						275	160	7-¾	8-¾

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 12—Spring-Loaded Pressure Relief Valves "P" Orifice (Effective Orifice Area = 6.38 square inches)

Materials(2)		Valve Size		ANSI Flange Class	Maximum Pressure (psig)									
					Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	4P6	150	150			285	185			285	80	7-¼	9
		4P6	300	150			285	285			285	80	7-¼	9
		4P6	300	150			525	525			285	150	8-¾	10
		4P6	600	150			1000	1000			285	150	8-¾	10
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	4P6	150	150				185	80		285	80	7-¼	9
		4P6	300	150				285	285		285	80	7-¼	9
		4P6	300	150				525	410		285	150	8-¾	10
		4P6	600	150				1000	825		285	150	8-¾	10
		4P6	900	150				1000	1000		285	150	8-¾	10
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	4P6	300	150					510	225	285	150	8-¾	10
		4P6	600	150					1000	445	285	150	8-¾	10
		4P6	900	150					1000	670	285	150	8-¾	10
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	4P6	150	150		275					275	80	7-¼	9
		4P6	300	150		275					275	80	7-¼	9
		4P6	300	150		525					275	150	8-¾	10
		4P6	600	150		1000					275	150	8-¾	10
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	4P6	150	150	175						175	80	7-¼	9
		4P6	300	150	175						175	80	7-¼	9
		4P6	300	150	300						275	150	8-¾	10
		4P6	600	150	480						275	150	8-¾	10

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 13—Spring-Loaded Pressure Relief Valves
"Q" Orifice (Effective Orifice Area = 11.05 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	6Q8	150	150			165	165			115	70	9-7/16	9-1/2
		6Q8	300	150			165	165			115	70	9-7/16	9-1/2
		6Q8	300	150			300	300			115	115	9-7/16	9-1/2
		6Q8	600	150			600	600			115	115	9-7/16	9-1/2
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	6Q8	150	150				165	80		115	70	9-7/16	9-1/2
		6Q8	300	150				165	165		115	70	9-7/16	9-1/2
		6Q8	300	150				300	300		115	115	9-7/16	9-1/2
		6Q8	600	150				600	600		115	115	9-7/16	9-1/2
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	6Q8	300	150					165	165	115	115	9-7/16	9-1/2
		6Q8	600	150					600	445	115	115	9-7/16	9-1/2
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	6Q8	150	150		165					115	70	9-7/16	9-1/2
		6Q8	300	150		165					115	70	9-7/16	9-1/2
		6Q8	300	150		300					115	115	9-7/16	9-1/2
		6Q8	600	150		600					115	115	9-7/16	9-1/2
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	6Q8	150	150	165						115	70	9-7/16	9-1/2
		6Q8	300	150	165						115	70	9-7/16	9-1/2
		6Q8	300	150	250						115	115	9-7/16	9-1/2
		6Q8	600	150	300						115	115	9-7/16	9-1/2

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 14—Spring-Loaded Pressure Relief Valves "R" Orifice (Effective Orifice Area = 16.00 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	6R8	150	150			100	100			60	60	9-7/16	9-½
		6R8	300	150			100	100			60	60	9-7/16	9-½
		6R10	300	150			230	230			100	100	9-7/16	10-½
		6R10	600	150			300	300			100	100	9-7/16	10-½
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	6R8	150	150				100	80		60	60	9-7/16	9-½
		6R8	300	150				100	100		60	60	9-7/16	9-½
		6R10	300	150				230	230		100	100	9-7/16	10-½
		6R10	600	150				300	300		100	100	9-7/16	10-½
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	6R8	300	150					100	100	60	60	9-7/16	9-½
		6R10	600	150					300	300	100	100	9-7/16	10-½
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	6R8	150	150			100				60	60	9-7/16	9-½
		6R8	300	150			100				60	60	9-7/16	9-½
		6R10	300	150			230				100	100	9-7/16	10-½
		6R10	600	150			300				100	100	9-7/16	10-½
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	6R8	150	150	55						55	55	9-7/16	9-½
		6R8	300	150	55						55	55	9-7/16	9-½
		6R10	300	150	150						100	100	9-7/16	10-½
		6R10	600	150	200						100	100	9-7/16	10-½

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 15—Spring-Loaded Pressure Relief Valves "T" Orifice (Effective Orifice Area = 26.00 square inches)

Materials(2)		Valve Size	ANSI Flange Class		Maximum Pressure (psig) Conventional and Balanced Bellows Valves						Conventional Valves	Bellows Valves	Center To Face Dimensions (inches)	
					Set Pressure Limit						Outlet Pressure Limit(1)			
Body/ Bonnet	Spring	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -76°F	-75°F to -21°F	-20°F to 100°F	450°F	800°F	1000°F	100°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 450°F Inclusive														
Carbon Steel	Carbon Steel	8T10	150	150			65	65			30	30	10-⅞	11
		8T10	300	150			65	65			30	30	10-⅞	11
		8T10	300	150			120	120			60	60	10-⅞	11
		8T10	300	150			300	300			100	100	10-⅞	11
Temperature Range, 451°F to 800°F Inclusive														
Carbon Steel	High Temperature Alloy Steel	8T10	150	150				65	65		30	30	10-⅞	11
		8T10	300	150				65	65		30	30	10-⅞	11
		8T10	300	150				120	120		60	60	10-⅞	11
		8T10	300	150				300	300		100	100	10-⅞	11
Temperature Range, 801°F to 1000°F Inclusive														
Chrome Molybdenum Steel	High Temperature Alloy Steel	8T10	300	150					120	120	60	60	10-⅞	11
		8T10	300	150					300	225	100	100	10-⅞	11
Temperature Range, -21°F to -75°F Inclusive														
Austenitic Stainless Steel	Carbon Steel	8T10	150	150		65					30	30	10-⅞	11
		8T10	300	150		65					30	30	10-⅞	11
		8T10	300	150		120					60	60	10-⅞	11
Temperature Range, -76°F to -450°F Inclusive														
Austenitic Stainless Steel	Low Temperature Alloy Steel	8T10	150	150	50						30	30	10-⅞	11
		8T10	300	150	50						30	30	10-⅞	11
		8T10	300	150	65						60	60	10-⅞	11

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 16—Pilot-Operated Pressure Relief Valves "D" Orifice (Effective Orifice Area = 0.110 square inch)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	1D2	150	150		285	170	80	285	4-1/8	4-1/2
	1D2	300	150		740	600	410	285	4-3/8	4-1/2
	1D2	600	150		1480	1200	825	285	4-3/8	4-1/2
	1D2	900	300		2220	1795		740	4-15/16	4-1/4
	1D2	1500	300		3705	2995		740	4-15/16	4-1/4
	1D2	2500	300		6170	4990		740	4-15/16	4-1/4
	1-1/2 D2	150	150		285	170	80	285	4-7/8	4-1/4
	1-1/2 D2	300	150		740	600	410	285	4-7/8	4-3/4
	1-1/2 D2	600	150		1480	1200	825	285	4-7/8	4-3/4
	1-1/2 D2	900	300		2220	1795		740	5-3/8	5-1/2
	1-1/2 D2	1500	300		3705	2995		740	5-3/8	5-1/2
	1-1/2 D2	2500	300		6170	4990		740	5-3/8	5-1/2
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	1D2	150	150	275				275	4-1/8	4-1/2
	1D2	300	150	720				275	4-3/8	4-1/2
	1D2	600	150	1440				275	4-3/8	4-1/2
	1D2	900	300	2160				720	4-15/16	4-3/4
	1D2	1500	300	3600				720	4-15/16	4-1/4
	1D2	2500	300	6000				720	4-15/16	4-3/4
	1-1/2 D2	150	150	275				275	4-7/8	4-1/4
	1-1/2 D2	300	150	720				275	4-7/8	4-3/4
	1-1/2 D2	600	150	1440				275	4-7/8	4-3/4
	1-1/2 D2	900	300	2160				720	5-3/8	5-1/2
	1-1/2D2	1500	300	3600				720	5-3/8	5-1/2
	1-1/2 D2	2500	300	6000				720	5-3/8	5-1/2

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 17—Pilot-Operated Pressure Relief Valves "E" Orifice (Effective Orifice Area = 0.196 square inch)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	1E2	150	150		285	170	80	285	4-⅛	4-½
	1E2	300	150		740	600	410	285	4-⅜	4-½
	1E2	600	150		1480	1200	825	285	4-⅜	4-½
	1E2	900	300		2220	1795		740	4-15/16	4-¾
	1E2	1500	300		3705	2995		740	4-15/16	4-¾
	1E2	2500	300		6170	4990		740	4-15/16	4-¾
	1-½E2	150	150		285	170	80	285	4-⅞	4-¾
	1-½E2	300	150		740	600	410	285	4-⅞	4-¾
	1-½E2	600	150		1480	1200	825	285	4-⅞	4-¾
	1-½E2	900	300		2220	1795		740	5-⅞	5-½
	1-½E2	1500	300		3705	2995		740	5-⅞	5-½
	1-½E2	2500	300		6170	4990		740	5-⅞	5-½
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	1E2	150	150	275				275	4-⅛	4-½
	1E2	300	150	720				275	4-⅜	4-½
	1E2	600	150	1440				275	4-⅜	4-½
	1E2	900	300	2160				720	4-15/16	4-¾
	1E2	1500	300	3600				720	4-15/16	4-¾
	1E2	2500	300	6000				720	4-15/16	4-¾
	1-½E2	150	150	275				275	4-⅞	4-¾
	1-½E2	300	150	720				275	4-⅞	4-¾
	1-½E2	600	150	1440				275	4-⅞	4-¾
	1-½E2	900	300	2160				720	5-⅞	5-½
	1-½E2	1500	300	3600				720	5-⅞	5-½
	1-½E2	2500	300	6000				720	5-⅞	5-½

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 18—Pilot-Operated Pressure Relief Valves "°F" Orifice (Effective Orifice Area = 0.307 square inch)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	1F2	150	150		285	170	80	285	4-1/8	4-1/2
	1F2	300	150		740	600	410	285	4-3/8	4-1/2
	1F2	600	150		1480	1200	825	285	4-3/8	4-1/2
	1F2	900	300		2220	1795		740	4-15/16	4-3/4
	1F2	1500	300		3705	2995		740	4-15/16	4-3/4
	1F2	2500	300		6170	4990		740	4-15/16	4-3/4
	1-½F2	150	150		285	170	80	285	4-7/8	4-1/4
	1-½F2	300	150		740	600	410	285	4-7/8	4-1/4
	1-½F2	600	150		1480	1200	825	285	4-7/8	4-1/4
	1-½F2	900	300		2220	1795		740	5-7/8	5-1/2
	1-½F2	1500	300		3705	2995		740	5-7/8	5-1/2
	1-½F2	2500	300		6170	4990		740	5-7/8	5-1/2
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	1F2	150	150	275				275	4-1/8	4-1/2
	1F2	300	150	720				275	4-3/8	4-1/2
	1F2	600	150	1440				275	4-1/8	4-1/2
	1F2	900	300	2160				720	4-15/16	4-3/4
	1F2	1500	300	3600				720	4-15/16	4-3/4
	1F2	2500	300	6000				720	4-15/16	4-3/4
	1-½F2	150	150	275				275	4-7/8	4-1/4
	1-½F2	300	150	720				275	4-7/8	4-1/4
	1-½F2	600	150	1440				275	4-7/8	4-1/4
	1-½F2	900	300	2160				720	5-7/8	5-1/2
	1-½F2	1500	300	3600				720	5-7/8	5-1/2
	1-½F2	2500	300	6000				720	5-7/8	5-1/2

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 19—Pilot-Operated Pressure Relief Valves "G" Orifice (Effective Orifice Area = 0.503 square inch)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)					Center to Face Dimensions (inches)	
				Set Pressure Limit				Outlet Pressure Limit(1)		
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
				Temperature Range, -20°F to 800°F						
Carbon Steel	1-½G3	150	150		285	170	80	285	5-¼	4-⅞
	1-½G3	300	150		740	600	410	285	5-¼	4-⅞
	1-½G3	600	150		1480	1200	825	285	5-¼	4-⅞
	1-½G3	900	300		2220	1795		740	6-⅜	6-¼
	1-½G3	1500	300		3705	2995		740	6-⅜	6-¼
	1-½G3	2500	300		6170	4990		740	6-⅜	6-¼
	2G3	150	150		285	170	80	285	5-⅜	4-⅞
	2G3	300	150		740	600	410	285	5-⅜	4-⅞
	2G3	600	150		1480	1200	825	285	5-⅜	4-⅞
	2G3	900	300		2220	1795		740	6-⅜	6-¼
	2G3	1500	300		3705	2995		740	6-⅜	6-¼
	2G3	2500	300		6170	4990		740	7	6-¼
				Temperature Range, -21°F to -450°F						
Austenitic Stainless Steel	1-½G3	150	150	275				275	5-¼	4-⅞
	1-½G3	300	150	720				275	5-¼	4-⅞
	1-½G3	600	150	1440				275	5-¼	4-⅞
	1-½G3	900	300	2160				720	6-⅜	6-¼
	1-½G3	1500	300	3600				720	6-⅜	6-¼
	1-½G3	2500	300	6000				720	6-⅜	6-¼
	2G3	150	150	275				275	5-⅜	4-⅞
	2G3	300	150	720				275	5-⅜	4-⅞
	2G3	600	150	1440				275	5-⅜	4-⅞
	2G3	900	300	2160				720	6-⅜	6-¼
	2G3	1500	300	3600				720	6-⅜	6-¼
	2G3	2500	300	6000				720	7	6-¼

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 20—Pilot-Operated Pressure Relief Valves "H" Orifice (Effective Orifice Area = 0.785 square inch)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	1-½H3	150	150		285	170	80	285	5-⅛	4-⅞
	1-½H3	300	150		740	600	410	285	5-⅛	4-⅞
	1-½H3	600	150		1480	1200	825	285	5-⅛	4-⅞
	1-½H3	900	300		2220	1795		740	6-⅛	6-¾
	1-½H3	1500	300		3705	2995		740	6-⅜	6-¾
	1-½H3	2500	300		6170	4990		740	6-⅜	6-¾
	2H3	150	150		285	170	80	285	5-⅛	4-⅞
	2H3	300	150		740	600	410	285	5-⅜	4-⅞
	2H3	600	150		1480	1200	825	285	5-⅜	4-⅞
	2H3	900	300		2220	1795		740	6-⅝	6-¾
	2H3	1500	300		3705	2995		740	6-⅝	6-¾
	2H3	2500	300		6170	4990		740	7	6-¾
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	1-½H3	150	150	275				275	5-⅛	4-⅞
	1-½H3	300	150	720				275	5-⅛	4-⅞
	1-½H3	600	150	1440				275	5-⅛	4-⅞
	1-½H3	900	300	2160				720	6-⅛	6-¾
	1-½H3	1500	300	3600				720	6-⅜	6-¾
	1-½H3	2500	300	6000				720	6-⅜	6-¾
	2H3	150	150	275				275	5-⅜	4-⅞
	2H3	300	150	720				275	5-⅜	4-⅞
	2H3	600	150	1440				275	5-⅜	4-⅞
	2H3	900	300	2160				720	6-⅝	6-¾
	2H3	1500	300	3600				720	6-⅝	6-¾
	2H3	2500	300	6000				720	7	6-¾

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 21—Pilot-Operated Pressure Relief Valves "J" Orifice (Effective Orifice Area = 1.287 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig) Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	2J3	150	150		285	170	80	285	5-⅛	4-⅞
	2J3	300	150		740	600	410	285	5-⅛	4-⅞
	2J3	600	150		1480	1200	825	285	5-⅛	4-⅞
	2J3	900	300		2220	1795		740	6-⅜	6-¾
	2J3	1500	300		3650	2995		740	6-⅜	6-¾
	2J3	1500	600		3705	2995		1480	6-⅜	6-¾
	2J3	2500	300		3650	3650		740	7	6-¾
	2J3	2500	600		6170	4990		1480	7	6-¾
	3J4	150	150		285	170	80	285	6-⅛	6-⅜
	3J4	300	150		740	600	410	285	6-⅛	6-⅜
	3J4	600	150		1480	1200	825	285	6-⅜	6-⅜
	3J4	900	300		2220	1795		740	7-½	7-⅛
	3J4	1500	300		3705	2995		740	7-½	7-⅛
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	2J3	150	150	275				275	5-⅛	4-⅞
	2J3	300	150	720				275	5-⅛	4-⅞
	2J3	600	150	1440				275	5-⅛	4-⅞
	2J3	900	300	2160				720	6-⅜	6-¾
	2J3	1500	300	3550				720	6-⅜	6-¾
	2J3	1500	600	3600				1440	6-⅜	6-¾
	2J3	2500	300	3550				720	7	6-¾
	2J3	2500	600	6000				1440	7	6-¾
	3J4	150	150	275				275	6-⅛	6-⅜
	3J4	300	150	720				275	6-⅛	6-⅜
	3J4	600	150	1440				275	6-⅜	6-⅜
	3J4	900	300	2160				720	7-½	7-⅛
	3J4	1500	300	3600				720	7-½	7-⅛

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 22—Pilot-Operated Pressure Relief Valves "K" Orifice (Effective Orifice Area = 1.838 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
	Inlet by Orifice by Outlet			Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body		Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	3K4	150	150		285	170	80	285	6-⅛	6-¾
	3K4	300	150		740	600	410	285	6-⅛	6-¾
	3K4	600	150		1480	1200	825	285	6-⅜	6-⅞
	3K4	900	300		2220	1795		740	7-½	7-⅛
	3K4	1500	300		3705	2995		740	7-½	7-⅛
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	3K4	150	150	275				275	6-⅛	6-¾
	3K4	300	150	720				275	6-⅛	6-¾
	3K4	600	150	1440				275	6-⅜	6-⅞
	3K4	900	300	2160				720	7-½	7-⅛
	3K4	1500	300	3600				720	7-½	7-⅛

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 23—Pilot-Operated Pressure Relief Valves "L" Orifice (Effective Orifice Area = 2.853 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	3L4	150	150		285	170	80	285	6-⅛	6-¾
	3L4	300	150		740	600	410	285	6-⅛	6-¾
	3L4	600	150		1240	1200	825	285	6-⅜	6-¾
	3L4	600	300		1480	1200	825	740	7-½	7-⅛
	3L4	900	300		2220	1795		740	7-½	7-⅛
	3L4	1500	300		2900	2900		740	7-½	7-⅛
	3L4	1500	600		3705	2995		1480	7-½	7-⅜
	4L6	150	150		285	170	80	285	7-¾	8-¼
	4L6	300	150		740	600	410	285	7-¾	8-¼
	4L6	600	150		1480	1200	825	285	7-¾	8-¼
	4L6	900	300		2220	1795		740	9-$^{13}/_{16}$	9-$^{3}/_{16}$
	4L6	1500	300		3705	2995		740	9-$^{13}/_{16}$	9-$^{3}/_{16}$
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	3L4	150	150	275				275	6-⅛	6-¾
	3L4	300	150	720				275	6-⅛	6-¾
	3L4	600	150	1200				275	6-⅜	6-¾
	3L4	600	300	1440				720	7-½	7-⅛
	3L4	900	300	2160				720	7-½	7-⅛
	3L4	1500	300	2825				720	7-½	7-⅛
	3L4	1500	600	3600				1440	7-½	7-⅜
	4L6	150	150	275				275	7-¾	8-¼
	4L6	300	150	720				275	7-¾	8-¼
	4L6	600	150	1440				275	7-¾	8-¼
	4L6	900	300	2160				720	9-$^{13}/_{16}$	9-$^{3}/_{16}$
	4L6	1500	300	3600				720	9-$^{13}/_{16}$	9-$^{3}/_{16}$

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 24—Pilot-Operated Pressure Relief Valves "M" Orifice (Effective Orifice Area = 3.60 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	4M6	150	150		285	170	80	285	7-¾	8-¼
	4M6	300	150		740	600	410	285	7-¾	8-¼
	4M6	600	150		1480	1200	825	285	7-¾	8-¼
	4M6	900	300		2220	1795		740	9-13/16	9-3/16
	4M6	1500	300		3705	2995		740	9-13/16	9-3/16
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	4M6	150	150	275				275	7-¾	8-¼
	4M6	300	150	720				275	7-¾	8-¼
	4M6	600	150	1440				275	7-¾	8-¼
	4M6	900	300	2160				720	9-13/16	9-3/16
	4M6	1500	300	3600				720	9-13/16	9-3/16

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 25—Pilot-Operated Pressure Relief Valves "N" Orifice (Effective Orifice Area = 4.34 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	4N6	150	150		285	170	80	285	7-¾	8-¼
	4N6	300	150		740	600	410	285	7-¾	8-¼
	4N6	600	150		1480	1200	825	285	7-¾	8-¼
	4N6	900	300		2220	1795		740	9-$^{13}/_{16}$	9-$^{3}/_{16}$
	4N6	1500	300		3705	2995		740	9-$^{13}/_{16}$	9-$^{3}/_{16}$
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	4N6	150	150	275				275	7-¾	8-¼
	4N6	300	150	720				275	7-¾	8-¼
	4N6	600	150	1440				275	7-¾	8-¼
	4N6	900	300	2160				720	9-$^{13}/_{16}$	9-$^{3}/_{16}$
	4N6	1500	300	3600				720	9-$^{13}/_{16}$	9-$^{3}/_{16}$

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 26—Pilot-Operated Pressure Relief Valves "P" Orifice (Effective Orifice Area = 6.38 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	4P6	150	150		285	170	80	285	7-¼	8-¼
	4P6	300	150		740	600	410	285	7-¼	8-¼
	4P6	600	150		1305	1200	825	285	7-¼	8-¼
	4P6	600	300		1480	1200	825	740	9-13/16	9-5/16
	4P6	900	300		2220	1795		740	9-13/16	9-5/16
	4P6	1500	300		3080	2995		740	9-13/16	9-5/16
	4P6	1500	600		3705	2995		1480	9-13/16	10-⅜
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	4P6	150	150	275				275	7-¼	8-¼
	4P6	300	150	720				275	7-¼	8-¼
	4P6	600	150	1260				275	7-¼	8-¼
	4P6	600	300	1440				720	9-13/16	9-5/16
	4P6	900	300	2160				720	9-13/16	9-5/16
	4P6	1500	300	2995				720	9-13/16	9-5/16
	4P6	1500	600	3600				1440	9-13/16	10-⅛

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 27—Pilot-Operated Pressure Relief Valves "Q" Orifice (Effective Orifice Area = 11.05 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	6Q8	150	150		285	170	80	285	9-7/16	9-½
	6Q8	300	150		740	600	410	285	9-7/16	9-½
	6Q8	600	150		1330	1200	825	285	9-11/16	9-½
	6Q8	600	300		1480	1200	825	740	9-11/16	10-7/16
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	6Q8	150	150	275				275	9-7/16	9-½
	6Q8	300	150	720				275	9-7/16	9-½
	6Q8	600	150	1285				275	9-11/16	9-½
	6Q8	600	300	1440				720	9-11/16	10-7/16

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 28—Pilot-Operated Pressure Relief Valves "R" Orifice (Effective Orifice Area = 16.00 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	6R8	150	150		285	170	80	285	9-7/16	9-1/2
	6R8	300	150		740	600	410	285	9-7/16	9-1/2
	6R8	600	150		915	915	825	285	9-11/16	9-1/2
	6R8	600	300		1480	1200	825	740	9-11/16	10-7/16
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	6R8	150	150	275				275	9-7/16	9-1/2
	6R8	300	150	720				275	9-7/16	9-1/2
	6R8	600	150	885				275	9-11/16	9-1/2
	6R8	600	300	1440				720	9-11/16	10-7/16

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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Table 29—Pilot-Operated Pressure Relief Valves "T" Orifice (Effective Orifice Area = 26.00 square inches)

Material(2)	Valve Size	ANSI Flange Class		Maximum Pressure (psig)						
				Set Pressure Limit				Outlet Pressure Limit(1)	Center to Face Dimensions (inches)	
Body	Inlet by Orifice by Outlet	Inlet	Outlet	-450°F to -21°F	-20°F to 100°F	500°F	800°F	100°F	Inlet	Outlet
Temperature Range, -20°F to 800°F										
Carbon Steel	8T10	150	150		285	170	80	285	10-¾	11
	8T10	300	150		740	600	410	285	10-¾	11
	8T10	600	150		900	900	825	285	11-1⅛	11
	8T10	600	300		1480	1200	825	740	11-1⅛	12
Temperature Range, -21°F to -450°F										
Austenitic Stainless Steel	8T10	150	150	275				275	10-¾	11
	8T10	300	150	720				275	10-¾	11
	8T10	600	150	870				275	11-1⅛	11
	8T10	600	300	1440				720	11-1⅛	12

Notes:
1. Outlet pressure limit for temperatures above 100°F shall not exceed the rating in ANSI/ASME B 16.34.
2. Materials given are minimum requirements for the pressure and temperature ratings. Other suitable materials may be used, as required for the service involved.

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APPENDIX A—
PRESSURE RELIEF VALVE SPECIFICATION SHEETS

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INSTRUCTIONS
SPRING-LOADED PRESSURE RELIEF VALVE SPECIFICATION SHEET

Line No.

1. Fill in item number.
2. Fill in user's pressure relief valve identification number.
3. Specify service, line, or equipment to be protected.
4. Specify number of valves required.
5. Specify the applicable Code(s) and whether Code Symbol nameplate stamping is required.
6. Valve should comply with API Standard 526.
7. Check fire or specify other basis of selection.
8. Specify whether a rupture disk is being used under the valve inlet.
9. Specify whether valve is conventional, balanced bellows, and/or balanced piston.
10. Give description of valve inlet (full nozzle, semi-nozzle, or other type).
11. Specify open or closed bonnet.
12. Specify metal-to-metal or resilient seat.
13. If other than API 527, specify seat test requirements.
14. Specify pipe size of inlet, flange rating, and type of facing.
15. Specify pipe size of outlet, flange rating, and type of facing.
16. Specify type of connection if other than flanged (e.g., threaded, socket weld, etc.).
17. Specify material of body.
18. Specify material of bonnet.
19. Specify material of seat or nozzle and disk.
20. If a resilient seat is required, specify material.
21. Specify material of guide.
22. Specify material of adjusting ring or rings.
23. Specify material of spring and spring washer.
24. Specify material of bellows.
25. Specify material of balanced piston.
26. Materials selected should be listed in NACE MR0175.
27. Specify any other special material requirements.
28. Specify screwed or bolted cap.
29. Specify if the valve is to have a plain or packed lifting lever or none.
30. Specify whether a test gag is required.
31. Specify whether a bug screen in the bonnet vent of a bellows or balanced piston valve is required.
32. Specify other accessories that are required (e.g., limit switch).
33. Indicate flowing fluid and state (liquid, gas, or vapor).
34. Specify quantity of fluid that the valve is required to relieve at relieving conditions and unit of measure (such as pounds per hr, gal per min, or cu ft per min).
35. Specify the molecular weight or specific gravity of the fluid at the flowing temperature.
36. Specify viscosity and unit of measure at the flowing temperature.
37. Specify operating pressure and unit of measure.
38. Specify set pressure and unit of measure.
39. Specify maximum blowdown as a percent of set pressure if different than manufacturer's standard.
40. Specify the latent heat of vaporization and unit of measure.
41. Specify the operating temperature and unit of measure.
42. Specify the actual temperature at relieving conditions and unit of measure.
43. Specify the increase in pressure in the discharge header as a result of flow.
44. Specify the amount of superimposed back pressure that normally exists on the valve outlet and unit of measure. If back pressure is variable, specify the minimum and maximum.
45. Specify the set pressure at which the valve is adjusted to open on the test stand. The cold differential test pressure includes corrections to the set pressure for the service conditions of back pressure or temperature or both.

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46. Specify the overpressure allowed, as a percent of set pressure or as a unit of measure.
47. Specify the compressibility factor, if used.
48. Give the specific heat ratio as $k_p = Cp/Cv$.
49. Specify the calculated orifice area, in square inches.
50. Specify the selected effective orifice area in square inches.
51. Specify the letter designation of the selected orifice.
52. Fill in the name of the manufacturer, if desired.
53. Fill in the manufacturer's model or type numbers, if desired.
54. Confirmation of orifice sizing calculations required from vendor.

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SPRING-LOADED PRESSURE RELIEF VALVE SPECIFICATION SHEET

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Requisition No.
Job No.
Date
Revised
By

GENERAL

1. Item Number:
2. Tag Number:
3. Service, Line, or Equipment Number:
4. Number Required:

VALVE DESIGN

9. Design Type:
   Conventional [ ] Bellows [ ] Balanced Piston [ ]
10. Nozzle Type: Full[ ] Semi [ ]
    Other [ ] Specify:
11. Bonnet Type: Open [ ] Closed [ ]
12. Seat Type: Metal to Metal [ ] Resilient [ ]
13. Seat Tightness: API 527 [ ]
    Other [ ] Specify:

CONNECTIONS

14. Inlet Size Rating Facing
15. Outlet Size Rating Facing
16. Other (Specify):

SERVICE CONDITIONS

33. Fluid and State:
34. Required Capacity Per Valve & Units:
35. Molecular Weight or Specific Gravity:
36. Viscosity at Flowing Temperature & Units:
37. Operating Pressure & Units:
38. Set Pressure & Units:
39. Blowdown: Standard [ ] Other [ ]
40. Latent Heat of Vaporization & Units:
41. Operating Temperature & Units:
42. Relieving Temperature & Units:
43. Built-up Back Pressure & Units:
44. Superimposed Back Pressure & Units
45. Cold Differential Test Pressure & Units:
46. Allowable Overpressure in Percent or Units:
47. Compressibility Factor, Z:
48. Ratio of Specific Heats:

BASIS OF SELECTION

5. Code: ASME VIII [ ]Stamp Req'd: Yes[ ] No[ ]
   Other [ ] Specify:
6. Comply With API 526: Yes[ ] No[ ]
7. Fire [ ] Other] [ ] Specify:
8. Rupture Disk: Yes[ ] No[ ]

MATERIALS

17. Body
18. Bonnet:
19. Seat (Nozzle): Disk:
20. Resilient Seat:
21. Guide
22. Adjusting Ring(s):
23. Spring: Washer:
24. Bellows:
25. Balanced Piston:
26. Comply With NACE MR0175: Yes [ ] No [ ]
27. Other (Specify):

ACCESSORIES

28. Cap: Screwed [ ] Bolted [ ]
29. Lifting Lever: Plain [ ] Packed [ ] None [ ]
30. Test Gag: Yes [ ] No [ ]
31. Bug Screen: Yes [ ] No [ ]
32. Other (Specify):

SIZING AND SELECTION

49. Calculated Orifice Area (in square inches):
50. Selected Orifice Area (in square inches):
51. Orifice Designation (letter):
52. Manufacturer:
53. Model Number:
54. Vendor Calculations Required: Yes [ ] No [ ]

Note: Indicate items to be filled in by the manufacturer with an asterisk (*).

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INSTRUCTIONS
PILOT-OPERATED PRESSURE RELIEF VALVE SPECIFICATION SHEET

Line No.

1. Fill in item number.
2. Fill in user's pressure relief valve identification number.
3. Specify service, line, or equipment to be protected.
4. Specify number of valves required.
5. Specify the applicable Code(s) and whether Code Symbol nameplate stamping is required.
6. Valve should comply with API Standard 526.
7. Check fire or specify other basis of selection.
8. Specify whether a rupture disk is being used under the valve inlet.
9. Specify type of main valve operation.
10. Specify number of pilots per main valve.
11. Specify if pilot is flow or non-flowing type.
12. Specify type of action, pop or modulating.
13. Specify sensing point as integral at main valve inlet or at a remote location.
14. Specify metal-to-metal or resilient seat.
15. If other than API 527, specify seat tightness test requirements.
16. Specify if pilot venting is to atmosphere, valve outlet, or other closed system.
17. Specify pipe size of inlet, flange rating, and type of facing.
18. Specify pipe size of outlet, flange rating, and type of facing.
19. Specify type of connection if other than flanges (e.g., threaded, socket weld, etc.).
20. Specify material of body.
21. Specify material of seat or nozzle and piston.
22. Specify material of resilient seat (if required) and seals.
23. Specify material of piston seal.
24. Specify material of piston liner or guide.
25. Specify material of diaphragm or bellows.
26. Specify material of pilot body and bonnet.
27. Specify material of pilot internals.
28. Specify material of seat and seals of the pilot.
29. Specify material of diaphragm.
30. Specify material of tubing and fittings.
31. Specify material of filter body and cartridge.
32. Specify material of spring.
33. Materials selected should be listed in NACE MR0175.
34. Specify any other special material requirements.
35. Specify if external filter is required.
36. Specify if valve is to have a plain or packed lifting lever or none.
37. Specify if field test connection is required.
38. Specify if field test indicator is required.
39. Specify if backflow preventer is required.
40. Specify if manual blowdown valve is required.
41. Specify if test gag is required.
42. Specify other accessories that are required.
43. Indicate flowing fluid and state (liquid, gas, or vapor).
44. Specify quantity of fluid that the valve is required to relieve at relieving conditions and unit of measure (such as pounds per hr, gal per min, or cu ft per min).
45. Specify the molecular weight or specific gravity of the fluid at the flowing temperature.
46. Specify viscosity and unit of measure at the flowing temperature.
47. Specify operating pressure and unit of measure.

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48. Specify set pressure and unit of measure.
49. Specify the maximum blowdown as a percent of set pressure if different than manufacturer's standard.
50. Specify the latent heat of vaporization and unit of measure.
51. Specify the operating temperature and unit of measure.
52. Specify the actual temperature at relieving conditions and unit of measure.
53. Specify the increase in pressure in the discharge header as a result of flow.
54. Specify the amount of superimposed back pressure that normally exists on the valve outlet and unit of measure. If back pressure is variable, specify the minimum and maximum.
55. Specify the set pressure at which the valve is adjusted to open on the test stand. The cold differential test pressure includes corrections to the set pressure for the service conditions of back pressure or temperature or both.
56. Specify the overpressure allowed, as a percent of set pressure or as a unit of measure.
57. Specify the compressibility factor, if used.
58. Give the specific heat ratio as $k_p = Cp/Cv$.
59. Specify the calculated orifice area, in square inches.
60. Specify the selected effective orifice area in square inches.
61. Specify the letter designation of the selected orifice.
62. Fill in the name of the manufacturer, if desired.
63. Fill in the manufacturer's model or type numbers, if desired.
64. Confirmation of orifice sizing calculations required from vendor.

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PILOT-OPERATED PRESSURE RELIEF VALVE SPECIFICATION SHEET

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Requisition No.
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Revised
By

GENERAL

1. Item Number:
2. Tag Number:
3. Service, Line, or Equipment Number:
4. Number Required:

VALVE DESIGN

9. Design Type: Piston [ ] Diaphragm [ ] Bellows [ ]
10. Number of Pilots:
11. Pilot Type: Flowing [ ] Non-Flowing [ ]
12. Pilot Action: Pop [ ] Modulating [ ]
13. Pilot Sense: Internal [ ] Remote [ ]
14. Seat Type: Metal to Metal [ ] Resilient [ ]
15. Seat Tightness: API 527 [ ]
    Other [ ] Specify:
16. Pilot Vent: Atmosphere [ ] Outlet [ ]
    Other [ ] Specify:

CONNECTIONS

17. Inlet Size Rating Facing
18. Outlet Size Rating Facing
19. Other (Specify):

SERVICE CONDITIONS

43. Fluid and State:
44. Required Capacity Per Valve & Units:
45. Molecular Weight or Specific Gravity:
46. Viscosity at Flowing Temperature & Units:
47. Operating Pressure & Units:
48. Set Pressure & Units:
49. Blowdown: Standard [ ] Other [ ]
50. Latent Heat of Vaporization & Units:
51. Operating Temperature & Units:
52. Relieving Temperature & Units:
53. Built-up Back Pressure & Units:
54. Superimposed Back Pressure & Units
55. Cold Differential Test Pressure & Units:
56. Allowable Overpressure in Percent or Units:
57. Compressibility Factor, Z:
58. Ratio of Specific Heats:

BASIS OF SELECTION

5. Code: ASME VIII [ ] Stamp Req'd: Yes[ ] No[ ]
   Other [ ] Specify:
6. Comply With API 526: Yes[ ] No[ ]
7. Fire [ ] Other [ ] Specify:
8. Rupture Disk: Yes[ ] No[ ]

MATERIALS, MAIN VALVE

20. Body:
21. Seat (Nozzle): Piston:
22. Resilient Seat: Seals:
23. Piston Seal:
24. Piston Liner/Guide:
25. Diaphragm/Bellows:

MATERIALS, PILOT

26. Body/Bonnet:
27. Internals:
28. Seat: Seals:
29. Diaphragm:
30. Tubing/Fittings:
31. Filter Body: Cartridge:
32. Spring:
33. Comply With NACE MR0175: Yes [ ] No [ ]
34. Other (Specify):

ACCESSORIES

35. External Filter: Yes [ ] No [ ]
36. Lifting Lever: Plain [ ] Packed [ ] None [ ]
37. Field Test Connection: Yes [ ] No [ ]
38. Field Test Indicator: Yes [ ] No [ ]
39. Backflow Preventer: Yes [ ] No [ ]
40. Manual Blowdown Valve: Yes [ ] No [ ]
41. Test Gag: Yes [ ] No [ ]
42. Other (Specify):

SIZING AND SELECTION

59. Calculated Orifice Area (in square inches):
60. Selected Orifice Area (in square inches):
61. Orifice Designation (letter):
62. Manufacturer:
63. Model Number:
64. Vendor Calculations Required: Yes [ ] No [ ]

Note: Indicate items to be filled in by the manufacturer with an asterisk (*).

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APPENDIX B—
PRESSURE RELIEF VALVE NAMEPLATE NOMENCLATURE

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Nomenclature	Description
Manufacturer's name or identifying trademark	Identification of manufacturer
Size	Nominal pipe size, inlet by outlet
Type, style, model, or figure no.	Manufacturer's designation
Orifice	Valve orifice size, standardized letter designations
Capacity at 10 percent overpressure	Pounds per hour of saturated steam, standard cubic feet per minute of air, at 60°F and 14.7 psia or U.S. gallons per minute of water at 70°F
Serial no. or shop no.	Manufacturer's identification
Set pressure, pounds per square inch gauge	Valve inlet pressure at which the pressure relief valve is adjusted to open under service conditions
Back pressure, pounds per square inch gauge	Constant (For example, 50 psig) Variable (For example, 0-50 psig)
Cold differential test pressure, pounds per square inch gauge (if applicable)	The pressure at which the pressure relief valve is adjusted to open on the test stand. The cold differential test pressure includes corrections to the set pressure for the service conditions of back pressure or temperature or both (See examples below)
Example 1—Conventional Valve	
Set pressure, psig	200
Back pressure, psig	Atmospheric (or 0)
Temperature, °F	400
Cold differential test pressure, psig	200 + manufacturer's recommended temperature correction
Example 2—Balanced Bellows Valve	
Set pressure, psig	200
Back pressure, psig	50, or 0-50
Temperature, °F	400
Cold differential test pressure, psig	200 + manufacturer's recommended temperature correction
Example 3—Conventional Valve	
Set pressure, psig	200
Back pressure, psig	50, constant superimposed
Temperature, °F	400
Cold differential test pressure, psig	200 – 50 + manufacturer's recommended temperature correction

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